

# Intro ML, X408 Final Project Evaluation Rubric

Solve using ML

## 1. Problem Definition (20 Points)

- Scope and Context (5 Points): The problem has been adequately scoped and the context has been clearly defined. **YES.**
- Objective (5 Points): The objective of the problem is clearly stated and is specific, measurable, achievable, relevant, and time-bound (SMART). **YES.**
- Data Identification (5 Points): The data needed to solve the problem has been identified and described. **YES.**
- Stakeholder Analysis (5 Points): The project team has identified all relevant stakeholders and taken into account their needs and constraints. **YES.**

## 2. Data Collection and Preparation (20 Points)

- Data Collection (5 Points): Data has been collected from relevant sources and has been thoroughly documented. **YES.**
- Data Cleaning (5 Points): The data has been cleaned and missing values, outliers, and inconsistencies have been addressed. **YES.**
- Data Preprocessing (5 Points): The data has been preprocessed appropriately, including techniques like scaling, normalization, encoding, etc. **YES.**
- Data Partitioning (5 Points): The data has been split into training, validation, and test sets in a reasonable manner. **YES.**

## 3. Model Selection and Development (20 Points)

- Model Choice (5 Points): The choice of machine learning model(s) is justified given the problem and the data. **YES.**
- Model Training (5 Points): The model has been trained effectively and performance on the training set is reported. **YES.**
- Model Validation (5 Points): The model has been validated on the validation set and performance metrics have been reported. **YES.**
- Model Optimization (5 Points): Hyperparameters have been tuned and any overfitting or underfitting has been addressed. **YES.**

## 4. Evaluation and Testing (20 Points)

- Performance Metrics (5 Points): The team has selected appropriate performance metrics for the problem and the model. **YES.**
- Testing (5 Points): The model has been tested on the test set and performance metrics have been reported. **YES.**
- Interpretation of Results (5 Points): The results have been interpreted in a meaningful way. **YES.**
- Comparison with Benchmarks (5 Points): The results are compared with relevant benchmarks or previous results. **YES.**

## 5. Communication and Presentation (20 Points). **Will take place on June 10, 2026.**

- Clarity (5 Points): The project and its results are clearly communicated.
- Visualizations (5 Points): Appropriate visualizations are used to help communicate the results.
- Documentation (5 Points): All code, data, and models are thoroughly documented.
- Teamwork (5 Points): The team worked together effectively and roles and responsibilities were distributed fairly.